

# Deep Learning for Evapotranspiration Estimation: A Systematic Review and LSTM-Based Case Study in Piedmont, Italy

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**Abstract**—This study investigates deep learning techniques employed to estimate Evapotranspiration ( $ET$ ), the process by which water is transferred from the soil and plants to the atmosphere through evaporation or transpiration. Performing accurate estimations of evapotranspiration is crucial in agriculture, where it can help to optimize irrigation and reduce water waste. This research reviews 159 papers according to prediction algorithms, performance metrics, datasets, and study areas, highlighting that tree based machine learning algorithms as Random Forest (RF) and temporal neural networks (RNN, LSTM) are the prediction instruments most frequently used, RMSE,  $R$  and  $R^2$  are the most used performance metrics, ground based datasets as FLUXNET are the most widely implemented, and Asian countries, mainly China and India, emerge as the most frequently studied. Additionally, a personal implementation of a Long Short-Term Memory (LSTM) model was developed and tested to perform temporal predictions of the actual evapotranspiration ( $AET$ ) in the Piedmont region in Italy. Moreover, the performances of the model were compared to other tests to understand its capabilities in cases where some climatological values are missing, and the transferability to generalize over different areas.

## I. INTRODUCTION

AGRICULTURE is a critical sector with crucial optimization needs to enhance crop yields and minimize water consumption. Evapotranspiration ( $ET$ ) is a key component of the water cycle, it quantifies the volume of water vapor transferred from the Earth's surface to the atmosphere through evaporation and transpiration in a specific period of time [1]. Therefore, accurate estimation of  $ET$  is essential for improving irrigation scheduling and helping in the early detection of drought conditions [2]. Additionally, in climate science and meteorology,  $ET$  plays a critical role in enhancing weather prediction models and monitoring the impacts of climate change [3].

However, directly measuring  $ET$  can be challenging because it necessitates the application of specific models that integrate various climatological and soil parameters.

Traditional  $ET$  estimation are primarily based on physical principles, such as the FAO-56 Penman-Monteith model, which integrates physiological characteristics to the FAO-24 Penman.

However, these methods frequently fail to capture the nonlinear relationships of the meteorological data and have significant limitations on performing  $ET$  estimations in absence of some of these data. For this reason, deep learning algorithms can be employed due to its ability to handle non-linear relationships and spot complex patterns [1].

## II. BACKGROUND

There are several different types of Evapotranspiration, each of these needs a different model to be estimated:

### A. Reference Evapotranspiration ( $ET_0$ )

Represents the evapotranspiration from a standardized reference surface, like well-watered grass, and it is used as a benchmark for estimating crop water requirements.

The most widely used technique to evaluate  $ET_0$  is the Penman-Monteith equation (FAO-56). This method depends on meteorological variables, as air temperature, relative humidity, solar radiation, and wind speed. Moreover, it does not require any local calibration, since it incorporates both physiological and aerodynamic parameters [4].

However, these climatological data are not always available in all the study areas. Therefore, alternative approaches are available and can be classified in three types, namely, radiation-based models (Priestley–Taylor), temperature-based models (Hargreaves and Samani, Trajkovic), and combination models (Penman) [5] [6] [7].

### B. Potential Evapotranspiration ( $PET$ )

Measures the maximum evapotranspiration possible from a specific surface, assuming unlimited water availability.

To evaluate  $PET$ , the same models used for  $ET_0$  can be applied, but with surface parameters adapted to the actual vegetation and land cover.

### C. Actual Evapotranspiration ( $AET$ )

Describes the real amount of water lost from a surface to the atmosphere, considering the limited water availability in the soil. Since it depends on water availability,  $AET$  is more complex to be estimated and measured over large areas.

Lysimeters, which are sensors that accurately quantify water loss by tracking variations in the water volume contained in the soil, can be used to directly measure  $AET$  in a specific location. Although this method is precise and reliable, it is also quite expensive [8].

Another direct measurement technique is the Eddy Covariance (EC) method, which captures water vapor fluxes above the canopy through high frequency measurements. EC provides detailed insight into evapotranspiration dynamics, but it requires sophisticated instrumentation and advanced data processing [9].

Other solutions, for broader spatial coverage and with less complexity, are based on satellite remote sensing [10]. Satellites collect various spectral indices that provides information

into vegetation condition, surface temperature, and energy balance components—such as NDVI, EVI, LST, LAI, SPI, and SPEI (see table I) [11]. These indices can be used as input to specific models that can measure the  $AET$ , such as the Surface Energy Balance (SEB) model [12]. Another commonly used approach is the crop coefficients method, which estimates  $AET$  by adjusting reference evapotranspiration  $ET_0$  with coefficients that account for the crop type, growth stage, and water stress conditions. Despite the wide implementation of this technique and its ease of application, a key limitation is that it cannot always be generalized over different areas without adjustment or calibration.

TABLE I  
DESCRIPTION OF SPECTRAL AND CLIMATIC INDICES USED FOR  $AET$  ESTIMATION

| Acronym | Full Name                                           |
|---------|-----------------------------------------------------|
| NDVI    | Normalized Difference Vegetation Index              |
| EVI     | Enhanced Vegetation Index                           |
| LST     | Land Surface Temperature                            |
| LAI     | Leaf Area Index                                     |
| SPI     | Standardized Precipitation Index                    |
| SPEI    | Standardized Precipitation Evapotranspiration Index |

### III. MACHINE LEARNING AND NEURAL NETWORK APPROACHES FOR EVAPOTRANSPIRATION ESTIMATION

Machine learning (ML) and neural networks (NN) are powerful tools for  $ET$  estimations due to their ability to handle complex, non-linear relationships in the data. Traditional  $ET$  estimation models often rely on meteorological parameters, that might be unavailable in some areas and can be costly to be measured. ML and NN approaches can address these limitations by performing pattern recognition, and accurate predictions even with incomplete data [13].

#### A. Traditional ML

1) *Random Forests (RF)*: Widely utilized for regression tasks, RF constructs an ensemble of decision trees, each trained on a randomly sampled subset of the training set. To compose the final prediction, the outputs of all individual trees are aggregated, typically by averaging the results. RF models are known for their strong generalization capabilities, robustness to outliers, and ease of hyperparameter optimization, making them effective for a wide range of prediction tasks [14].

2) *Support Vector Machine (SVM)*: Transforms the input vector into an higher-dimensional feature space, and finds the optimal hyperplane that separate these points. Generally, SVM can be mathematically explained with the Eq. 1

$$Y = w\varphi(X) + b \quad (1)$$

where  $Y$  is the output vector,  $w$  is the weight vector,  $\varphi$  is the kernel function,  $X$  is the input vector, and  $b$  is the bias

term [15].

3) *Gradient Boosting Machine (GBM)*: Ensemble of decision trees, where each tree has the aim to correct the errors made by the previous and improve the final prediction. XGBoost (Extreme gradient boosting) and LightGBM (Light gradient boosting machine) models are newer version that improves the overall performance of the traditional GBM [14].

#### B. Neural Networks

1) *Multilayer Perceptrons (MLP)*: Composed by layers of connected nodes (neurons) that transform inputs through weighted connections and non-linear activation functions. During the training, MLP adjust the weights to learn to approximate complex relationships between input features.

2) *Recurrent Neural Network (RNN)*: A version of the standard artificial neural network (ANN) that process sequential data by maintaining a memory of previous inputs through internal states. RNNs are used for modeling time series data, since their ability to learn temporal dependencies.

3) *Long Short-Term Memory (LSTM)*: Enhanced type of RNN designed to capture long-term dependencies in sequential data. LSTM consists of layers of memory cells, each cell contains three gates: forget, input, and output. The forget gate controls the information coming from the previous state that will no longer be stored, the input gate specifies the input features that will be stored in the next state, and the output gate specifies the information from the new state that will be output. This gated architecture enables LSTMs to effectively retain relevant information over long sequences, making them particularly suitable for modeling temporal patterns [3].

4) *Convolutional Neural Network (CNN)*: Evolution of standard artificial neural network (ANN), CNNs are distinguished by their use of convolutional layers inspired by the mathematical operation of convolution. These layers filter the input by applying a mask of learnable weights to extract features and detect local patterns. In 1D CNNs, this mechanism can be applied to analyze time series data and extract meaningful features representing temporal variations. Moreover, 2D CNNs are widely used for image processing tasks due to their ability to identify spatial hierarchies and complex visual patterns [1].

### IV. STATE OF THE ART

The review focused on the classification of recent research on the field of machine learning (ML) and deep learning (DL) usage for evapotranspiration ( $ET$ ) estimation. With the goal of highlight the most commonly used approaches, 159 papers have been classified according to the model applied, performance metrics adopted, dataset used, and geographic areas covered.

### A. Machine Learning and Deep Learning Models for $ET$ Estimation

As shown in the figure 1, the classification reveals that Recurrent Neural Networks (RNN) models are the most frequently employed for  $ET$  estimation, since their strong capability of understanding temporal dependencies and seasonal patterns of the meteorological data that affects  $ET$  values. This category includes several variants such as BiRNN, LSTM, BiLSTM, ConvLSTM, GRU, and BiGRU. Among these, the Long Short-Term Memory (LSTM) architecture is the most widely adopted, appearing in 75 studies (Table II) [16] [17] [3] [18].

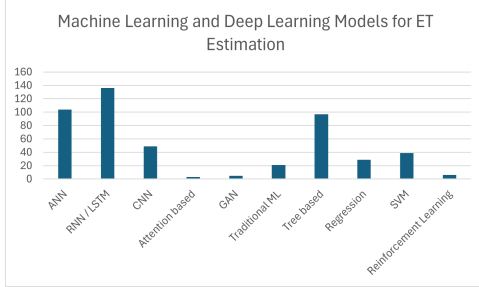


Fig. 1. Summary of algorithms used in the reviewed studies

TABLE II  
RNN-BASED ALGORITHMS USED IN THE REVIEWED STUDIES

| Algorithm | No. of Studies | Algorithm | No. of Studies |
|-----------|----------------|-----------|----------------|
| RNN       | 14             | BiGRU     | 3              |
| BiRNN     | 2              | SSR-LSTM  | 3              |
| LSTM      | 75             | ConvLSTM  | 5              |
| BiLSTM    | 18             | NARX      | 1              |
| GRU       | 15             |           |                |

Other popular solutions are the standard Artificial Neural Networks (ANNs) (Table III), such as the Multi-Layer Perceptron (MLP) and Deep Belief Networks (DBN) [19] [20] [21], as well as Tree based machine learning models like Random Forests (RF), M5Tree, and Gradient Boosting methods, including its optimized variants LightGBM and XGBoost (Table IV); see Figure 1 [22] [23] [24]. These models can be useful for understanding complex patterns in the climatological spatial data and their relationship with evapotranspiration  $ET$  values.

TABLE III  
ANN-BASED ALGORITHMS USED IN THE REVIEWED STUDIES

| Algorithm | No. of Studies | Algorithm | No. of Studies |
|-----------|----------------|-----------|----------------|
| DNN       | 52             | DBN       | 7              |
| MLP       | 6              | DL        | 34             |
| DeepFM    | 1              | GRNN      | 2              |
| BPNN      | 2              |           |                |

TABLE IV  
TREE BASED ALGORITHMS USED IN THE REVIEWED STUDIES

| Algorithm | No. of Studies | Algorithm | No. of Studies |
|-----------|----------------|-----------|----------------|
| DT        | 5              | GB        | 15             |
| CART      | 1              | XGB       | 14             |
| BCART     | 1              | LightGB   | 4              |
| M5Tree    | 5              | AdaBoost  | 1              |
| RF        | 45             | DRF       | 1              |
| BT        | 1              |           |                |

### B. Performance Metrics Used

Performance metrics were classified considering how they evaluate the goodness of the predictions.

Figure 2 illustrates with an histogram the occurrences of each class of metrics in the reviewed papers. The most common class is the Squared Error metrics, which includes Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Root Mean Square Deviation (RMSD); see Table V. These metrics penalize significant deviations between the predicted  $ET$  value and the real one [1] [22] [25].

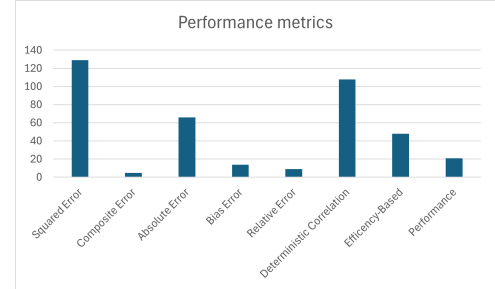


Fig. 2. Summary of performance metrics used in the reviewed studies

TABLE V  
SQUARED ERROR METRICS USED IN THE REVIEWED STUDIES

| Metric | No. of Studies |
|--------|----------------|
| RMSE   | 100            |
| MSE    | 28             |
| RMSD   | 1              |

Another widely implemented class of metrics is the Deterministic Correlation, which refers to the correlation coefficient ( $R$ ), the coefficient of determination ( $R^2$ ), and Variance Accounted For (VAF); see Table VI.

These metrics are widely implemented to evaluate how well the predicted trends match the observed patterns [26] [23].

TABLE VI  
DETERMINISTIC CORRELATION METRICS USED IN THE REVIEWED STUDIES

| Metric | No. of Studies |
|--------|----------------|
| $R$    | 21             |
| $R^2$  | 85             |
| VAF    | 1              |

Additionally, several studies employed Absolute Error metrics, such as Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE), since their interpretability and robustness to outliers (see Table VII) [27] [28] [29].

TABLE VII  
EFFICIENCY-BASED METRICS USED IN THE REVIEWED STUDIES

| Algorithm | No. of Studies | Algorithm | No. of Studies |
|-----------|----------------|-----------|----------------|
| MAE       | 52             | MADE      | 1              |
| MAPE      | 9              | MdAPE     | 3              |
| WAPE      | 1              | AARE      | 1              |
| MAD       | 1              |           |                |

Lastly, Efficiency-Based metrics, as the Nash–Sutcliffe Efficiency (NSE), the Kling–Gupta Efficiency (KGE), and the Index of Agreement (IOA), were also frequently used due to their ability to assess how well the model predicts relative to the mean of the observed data; see Table VIII [30] [31] [32].

TABLE VIII  
EFFICIENCY-BASED METRICS USED IN THE REVIEWED STUDIES

| Algorithm | No. of Studies | Algorithm | No. of Studies |
|-----------|----------------|-----------|----------------|
| NSE       | 38             | NNSE      | 1              |
| KGE       | 6              | IOA       | 3              |

### C. Input Data Sources

*ET* prediction relies on a combination of meteorological data (e.g., temperature, humidity, solar radiation), ground measurements (e.g., water, energy, gas flux), and satellite derived surface indices (e.g., NDVI, EVI, LST).

Figure 3 shows that the primary source of data comes from ground based measurements and weather stations.

This source of data provide high frequency accurate data, but their spatial coverage is limited to the locations where the measurements are taken, for this reason they are most useful in studies where the coverage area is local or regional (see figure 4 and figure 5).

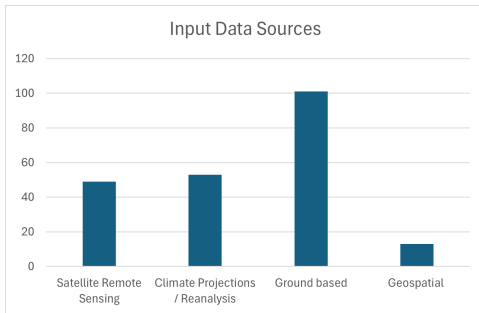


Fig. 3. Summary of data sources used in the reviewed studies

1) *Ground based measurements*: 101 papers over 159 have used ground based data to feed their prediction algorithms. Between them, the most commonly used dataset is FLUXNET, a global network of meteorological tower sites distributed worldwide, where each uses the eddy covariance technique to

measure exchanges of carbon dioxide, water vapor, and energy between terrestrial ecosystems and the atmosphere [26] [33] [34] [35].

Weather stations collect meteorological data, mostly used to feed traditional reference methods (e.g. FAO-56 Penman-Monteith, Hargreaves-Samani) and provide standard *ET* estimates [28] [36].

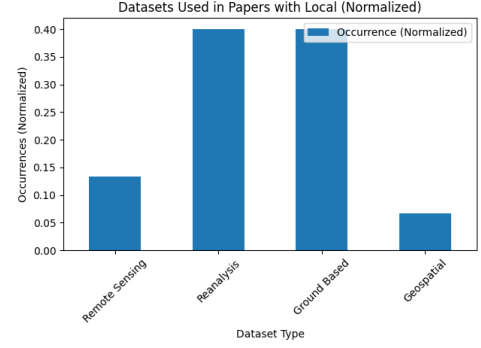


Fig. 4. Datasets Used in Papers with Local study areas

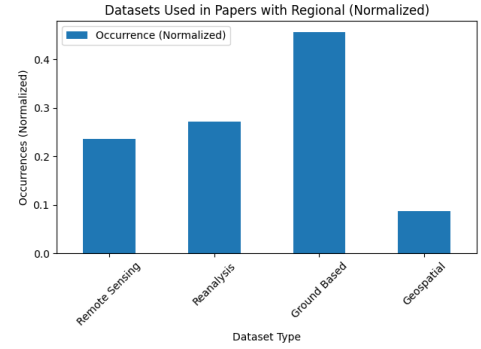


Fig. 5. Datasets Used in Papers with Regional study areas

In studies with a global study area, the most common choice is to use remote sensing based datasets or reanalysis datasets (see figure 6).

2) *Satellite remote sensing datasets*: Satellite products such as MODIS (Moderate Resolution Imaging Spectroradiometer), Copernicus, GRACE (Gravity Recovery and Climate Experiment), HyTES, and GLASS provide indices related to land surface temperature, vegetation coverage, and water storage changes. These datasets enable *ET* estimation over large areas but often at lower temporal or spatial resolutions [12] [37] [38] [39].

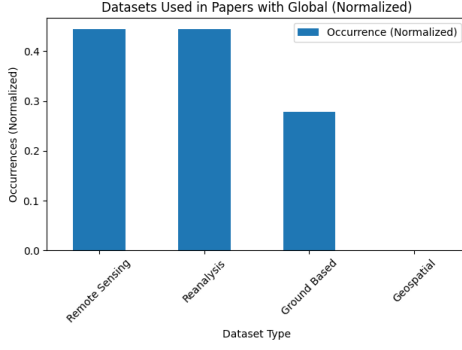


Fig. 6. Datasets Used in Papers with Global study areas

When the study has a long-term projection, is important to rely on consistent, temporally extensive, and spatially comprehensive data of atmospheric and surface conditions. For these studies the most convenient choice is to use climate model projections or reanalysis datasets (see figure 7).

3) *Climate model projections and reanalysis datasets:* Climate models, as CMIP6 (Coupled Model Intercomparison Project Phase 6), simulate future climate scenarios and provide projections of meteorological variables influencing *ET*. Reanalysis datasets such as ERA5, GLDAS (Global Land Data Assimilation System), and CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) integrate satellite observations and model outputs to produce consistent, gridded climate and hydrological variables over extended periods [40] [41] [42].

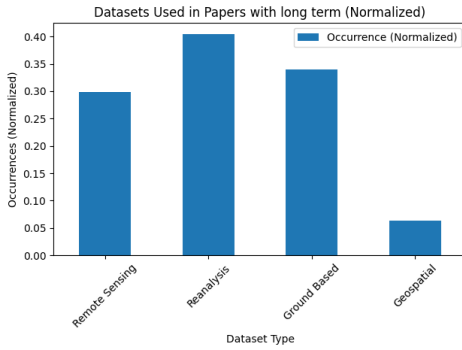


Fig. 7. Datasets Used in Papers with Long Term Projection

For local and regional studies, geospatial datasets could be a suitable choice since they provide accurate information directly referred to a specific area.

4) *Geospatial datasets:* In this review only few studies have been found using these datasets. The reason is that, even if they are reliable and consistent, they are available only for some specific areas. For example, some of these datasets found during the review are CAMELS (U.S. river basins), and CANOPEX (Canadian river basins) [43] [44].

## D. Geographic and Temporal Distribution

The review highlights that the majority of evapotranspiration studies are located in Asia (see figure 8).

Particularly, the Asian countries that appeared the most are China and India, due to their vast agricultural regions stressed by the high demand and by the different climatological conditions [22] [45] [46] [47].

Outside the Asiatic continent, other popular countries found during the review are the United States and Australia [48] [49]. Both cover a wide geographic area characterized by significant climatic variability and high agricultural water requirements. Only few studies have been found focusing over European and Mediterranean regions. The reason is that in these regions is difficult to perform *ET* prediction due to several environmental factors, as the intense seasonal droughts, fragmented land cover, and heterogeneous climatic patterns.

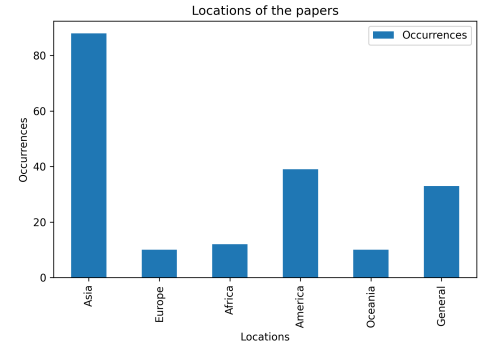


Fig. 8. Summary of locations of the studies

## E. Preprocessing and Feature Engineering Techniques

A common issue in *ET* prediction datasets used to train a neural network, is the presence of relationships among the input variables, that can cause the model to overfit and have bad prediction performances. These datasets contains directly related features as meteorological values, vegetation indices and surface attributes.

Therefore, it is common to reduce collinearity, presence of linear relationships among input variables, using preprocessing algorithms such as Principal Component Analysis (PCA). Independent Component Analysis (ICA) may also be used to extract statistically independent components [50] [51]. Other relevant algorithms found are the PSO (Particles Swarm Optimization), used to find the best hyperparameters for the neural network, and SHAP (SHapley Additive exPlanations) that interprets model predictions and the importance of each feature [52].

## V. MATERIALS AND METHODS

After the review, a personal implementation of a neural network has been developed and tested to perform *ET* predictions.

The goal was to use all the knowledge obtained during the review to implement a reliable and efficient model to predict the actual evapotranspiration (*AET*) over the Piedmont region in Italy.

### A. Dataset and Study Area

The development of the neural network started by finding the best dataset to perform *AET* estimations in Italy. As the review shows, few studies have implemented evapotranspiration model in Italy due to the climatological complexity of the mediterranean area.

Moreover, evapotranspiration (*ET*) trend in the Piedmont region, located in the northern Italy, is characterized by high spatial and temporal variability influenced by its position between Mediterranean warm areas and more temperate alpine zones [53] [54]. To address this challenge, two preprocessed datasets have been used, the first contains climatological variables used to perform regression, and the second for the target *AET* values. These two dataset have been aligned in space and time, to then obtain an unique dataset that for each pixel (longitude, latitude) of Italy contains 8 climatological features (minimum, average, and maximum air temperature, minimum and maximum relative humidity, wind speed, surface solar radiation, precipitation) and the related *AET* value for each day. The study focused only on the positions of the Piedmont region and in a time period that elapse from 01-01-2018 to 01-01-2022.

1) *MADIA - Meteorological variables for agriculture: A dataset for the Italian area*: This dataset provides detailed agrometeorological variables across Italy, gridded at a  $0.25^\circ$  resolution, and derived from ERA5 reanalysis data for the period 1981–2022.

It contains daily time series of minimum, average, and maximum air temperature, minimum and maximum relative humidity, wind speed, surface solar radiation, precipitation, and the reference evapotranspiration ( $ET_0$ ) calculated using the FAO Penman-Monteith method.

The MADIA dataset is owned and maintained by the Council for Agricultural Research and Economics (CREA), an Italian public research institution focused on agriculture and environment.

The dataset is described and available in the paper [55].

2) *GLEAM (Global Land Evaporation Amsterdam Model)*: GLEAM is a satellite based framework that provides global, daily estimates of terrestrial evapotranspiration (*ET*) and its components at  $0.25^\circ$  resolution.

It uses the Priestley–Taylor equation (or Penman in newer versions) to calculate the potential evapotranspiration (*PET*), which estimates the maximum possible *ET* assuming unlimited water availability.

It adjusts *PET* with an evaporative stress factor based on vegetation optical depth (VOD), an index for vegetation water content, and a multi-layer soil moisture model. GLEAM outputs actual evapotranspiration *AET* by summing evaporation and transpiration components. Data are available at <https://www.gleam.eu/> (accessed May 25, 2025), where registered users can download data via SFTP.

3) *Datasets Alignment*: MADIA dataset was downloaded as a zip folder from <https://zenodo.org/records/7252361> (accessed May 27, 2025), as referenced in the related publication

[55]. The folder contains a CSV file for each year from 1981 to 2022, and each file provides daily records for multiple spatial grid points across Italy, with 8 meteorological features for each date and location (see Table IX).

Since this study focuses exclusively on the Piedmont region, a spatial filter was applied to select only the grid points located within the area bounded by latitudes  $44.0^\circ$  to  $46.5^\circ$  and longitudes  $6.5^\circ$  to  $9.0^\circ$ .

Figure 9 displays all the spatial points available in the dataset, highlighting in red those corresponding to the Piedmont region. Additionally, the dataset was temporally filtered to include data only from 2018 to 2022.

TABLE IX  
MADIA DATASET DESCRIPTION

| Acronym   | Description                                                     |
|-----------|-----------------------------------------------------------------|
| longitude | longitude coordinate                                            |
| latitude  | latitude coordinate                                             |
| time      | date of the day                                                 |
| tasmin    | mean of daily minimum near-surface air temperature              |
| tasmax    | mean of daily maximum near-surface air temperature              |
| tasmean   | mean of daily average near-surface air temperature              |
| rhmin     | mean of daily minimum near-surface relative air humidity        |
| rhmax     | mean of daily maximum near-surface relative air humidity        |
| ws10      | mean of daily wind speed                                        |
| ssrd      | mean of daily surface solar radiation (shortwave radiation)     |
| ppn       | sum of daily water precipitation                                |
| pev       | sum of daily crop $ET_0$ estimated by FAO Penman-Monteith       |
| expver    | code which identifies temporary data when expver=5              |
| mask      | boolean code to identify cells belonging to the Italian country |

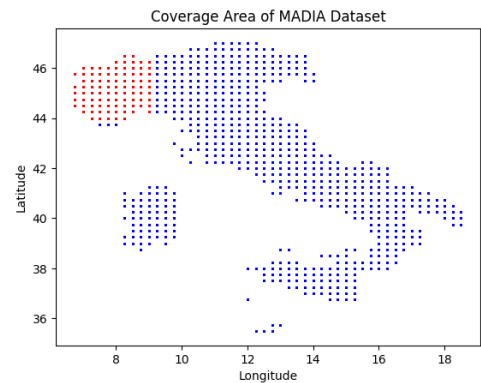


Fig. 9. Spatial Points of the filtered MADIA Dataset

GLEAM dataset was downloaded as annual NetCDF (NC) files through SFTP connection, each containing daily values for a specific variable. For this study, the NC files corresponding to daily *AET* values from 2018 to 2022 were downloaded. However, since the downloaded NC files had a global spatial coverage, the same spatial filter used for MADIA dataset was applied to extract only the grid points related to Piedmont.

The final dataset consists of 1,826 days (five years, including one leap year) and a spatial grid of 25 latitude points by 22 longitude points. This results in 550 pixels per day ( $25 \times 22$ ). Figure 10 shows the *AET* values contained in the extracted grid points over the Piedmont region in the first day of the dataset. Red pixels (14 in total) highlights Null values corresponding to the sea.

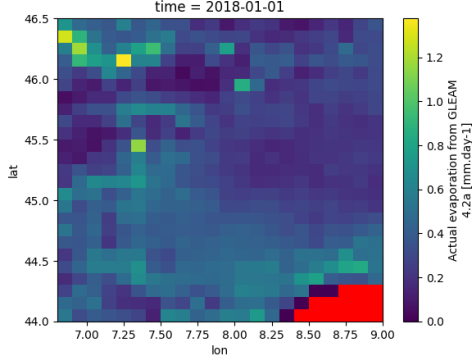


Fig. 10. Spatial Points of the Filtered GLEAM Dataset

Although the two datasets refer to the same geographic area, their spatial points are not perfectly aligned. To construct an unified dataset, each pixel in the MADIA dataset was matched, for each day, with the *AET* value of the nearest spatial point in the GLEAM dataset. The resulting dataset can be seen as the MADIA dataset with a new feature added, which represents the daily *AET* value for each pixel. Table X summarize the minimum, average, and maximum spatial shifts applied during the alignment.

TABLE X  
MINIMUM, MEAN, AND MAXIMUM SPATIAL SHIFT (IN DEGREES)  
APPLIED DURING DATASET ALIGNMENT

| Distance | Degrees |
|----------|---------|
| Min.     | 0.0     |
| Max.     | 0.112   |
| Mean     | 0.049   |

Figure 11 illustrates the final dataset, composed by the same grid points of the MADIA dataset but with the associated *AET* value.

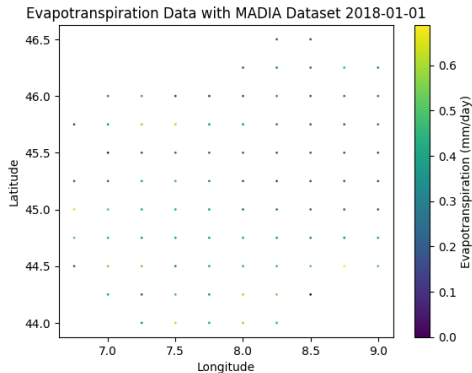


Fig. 11. Spatial Points of the Final Dataset

## B. Data Preprocessing

Before training the neural network, several preprocessing steps were applied to enhance data quality, reduce redundancy, and address missing values.

1) *Handling Missing Values*: The dataset may contain missing (Null) values in the *AET* field, particularly in pixels located near coastal areas or where GLEAM does not provide data. To ensure consistent input across the entire study area, all records with missing *AET* values were removed from the dataset (see Table XI).

TABLE XI  
SAMPLE COUNT BEFORE AND AFTER REMOVING MISSING *AET* VALUES

| Dataset Status                 | Number of Samples |
|--------------------------------|-------------------|
| Before Removing Missing Values | 160,688           |
| After Removing Missing Values  | 158,862           |

2) *Normalization*: To prepare a dataset to feed a neural network, normalization is a crucial step to ensure equal contribution of each feature during the learning process, and improves convergence time during training.

In this study, both the input features and the target variable (*AET*) were normalized using the Standard Scaler. This method transforms each feature to have zero mean and unit variance according to the formula 2:

$$x' = \frac{x - \mu}{\sigma} \quad (2)$$

where:

- $x$  is the original feature value,
- $\mu$  is the mean of the feature,
- $\sigma$  is the standard deviation of the feature.

The final set of input features and the target variable used in the regression task is listed in Table XII.

TABLE XII  
FINAL DATASET STRUCTURE

| Feature     | Description                                          |
|-------------|------------------------------------------------------|
| day_of_year | Integer (1–365/366)                                  |
| tasmin      | Daily minimum temperature [°C]                       |
| tasmean     | Daily mean temperature [°C]                          |
| tasmax      | Daily maximum temperature [°C]                       |
| rhmin       | Daily minimum relative humidity [%]                  |
| rhmax       | Daily maximum relative humidity [%]                  |
| ws10        | Wind speed at 10 meters [m/s]                        |
| ssrd        | Surface solar radiation [W/m²]                       |
| ppn         | Precipitation [mm/day]                               |
| E           | Actual Evapotranspiration [mm/day] (target variable) |

3) *Feature Correlation Analysis*: A first experiment has revealed an unexpected behavior during the training phase of the neural network, the loss function (evaluated using both Mean Squared Error and Huber loss) dropped to very small values after just a single epoch, indicating potential overfitting or a simplistic learning scenario.

To investigate the cause, a correlation analysis was performed using a correlation matrix (see Figure 12). This matrix

measures the linear relationships between pairs of features, where values close to  $\pm 1$  indicate strong positive or negative correlations that can cause the network to redundantly learn the same patterns and so overfit.

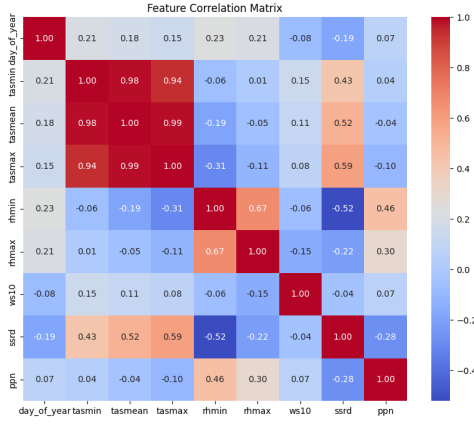


Fig. 12. Features Correlation Matrix

To address this issue, Principal Component Analysis (PCA) algorithm has been used to transform the original set of correlated features into a new set of uncorrelated components called principal components (see Table XIII).

TABLE XIII  
SAMPLE COUNT BEFORE AND AFTER REMOVING MISSING *AET* VALUES

| Dataset Status | Number of Features |
|----------------|--------------------|
| Before PCA     | 9                  |
| After PCA      | 6                  |

### C. Neural Network Architecture

Training was conducted exclusively on temporal trends to make the model learn how to predict the *AET* of the following day based only on the climatological evolution of the previous days, without considering the locations and the data of the next day.

To perform this task, a Long Short-Term Memory (LSTM) network was developed, a type of Recurrent Neural Network (RNN) particularly effective in modeling temporal dependencies and sequential patterns. The LSTM architecture was developed in Python by using the TensorFlow library, and the source code is publicly available at <https://github.com/CavaWasTaken/InnovativeWirelessPlatformsForIoT>.

The final network is composed by five layers, totaling 122,753 trainable parameters:

- **First Layer:** An LSTM layer that receives a batch of sequences, each consisting of 30 days of climatological data (sequence length defined as a hyperparameter). The features of each day in the sequence is processed by an LSTM cell, and each cell (except the first) also receives input from the previous cell to model temporal dependencies. This layer uses 128 units and is configured with `return_sequences=True`, meaning that its output will be composed by the outputs of each cell, and its shape is `(batch_size, 30, 128)`.

- **Second Layer:** A second LSTM layer that processes the sequence output from the first layer. This layer uses 64 units and has `return_sequences=False`, so it returns only the output of the final cell, with shape `(batch_size, 64)`.
- **Third Layer:** A Dropout layer that randomly disables 10% of the neurons during training to help prevent overfitting.
- **Fourth Layer:** A Dense (fully connected) layer with 64 units and ReLU activation, which transforms the input from the previous layer into a higher-level representation.
- **Fifth Layer:** A final Dense layer with 1 unit, which performs the regression task and outputs the predicted *AET* value.

The full architecture is illustrated in Figure 13.

| Layer (type)                          | Output Shape    | Param # |
|---------------------------------------|-----------------|---------|
| lstm (LSTM)                           | (None, 30, 128) | 65,120  |
| lstm_1 (LSTM)                         | (None, 64)      | 49,408  |
| dropout (Dropout)                     | (None, 64)      | 0       |
| dense (dense)                         | (None, 64)      | 4,160   |
| dense_1 (dense)                       | (None, 1)       | 63      |
| Total params: 122,753 (479.58 KB)     |                 |         |
| Trainable params: 122,753 (479.58 KB) |                 |         |
| Non-trainable params: 0 (0.00 B)      |                 |         |

Fig. 13. LSTM Architecture

## VI. EXPERIMENTS AND EVALUATION

### A. Sequence Construction

Since LSTM model requires to receive as input sequences of data, the dataset was segmented into sequences of 30 consecutive days of daily climatological variables, and each sequence was paired with the *AET* value of the following day.

The sequences were randomly shuffled to prevent the model from overfitting seasonal trends. An 80%–20% train-test split was applied to the sequences (see Table XIV).

TABLE XIV  
TRAIN AND TEST SET SPLITTING

| Set      | Features Shape   | Target Variable Shape |
|----------|------------------|-----------------------|
| Original | (156,252; 30; 6) | (156,252;)            |
| Train    | (125,001; 30; 6) | (125,001;)            |
| Test     | (31,251; 30; 6)  | (31,251;)             |

### B. Training Phase

The model was trained over 50 epochs using a batch size of 64 sequences. A validation test of 20% of the training data was used to monitor the model's performance during the training phase.

The optimization was performed using the Adam optimizer and the Mean Squared Error (MSE) was used as the loss function. Each epoch involved 1563 iterations and required approximately 18 seconds, totaling 932.28 seconds for the full

training phase. The complexity of the model required the usage of Google Colab GPU capabilities to perform the training.

### C. Testing Phase and Results

Testing was performed on the remaining 20% of the dataset. Since the model was trained on normalized data, the predicted *AET* values needs to be rescaled to their original range for better understanding.

Figure 14 shows a scatter plot comparing the predicted *AET* values to the true values. The red line indicates the ideal 1:1 regression line, helping visualize the goodness of the test.

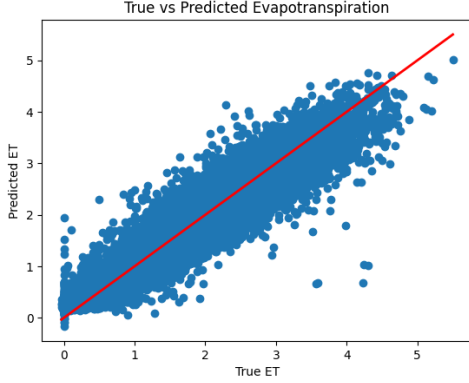


Fig. 14. LSTM Regression Predictions vs. Ground Truth

Three metrics were used to assess prediction performance:

- **$R^2$  (Coefficient of Determination)** – Can take values from 0 to 1, closer is to 1 and more the predicted values replicate the trend of the ground truth values.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

- **RMSE (Root Mean Squared Error)** – Evaluates the amount of error in the same units as the target variable.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

- **MAE (Mean Absolute Error)** – Represents the average absolute difference between predicted and true values.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

The values obtained from the evaluation metrics offer a clear view of the model's performance.  $R^2$  scores a value of 0.9392, indicating that nearly 94% of the variance in the actual *AET* values is explained by the model's predictions. This confirms that the model has learned to capture the underlying temporal patterns effectively.

The RMSE of 0.2697 [mm/day] and MAE of 0.2000 [mm/day] indicate consistent and relatively small prediction errors across the dataset.

The results are summarized in Table XV.

TABLE XV  
EVALUATION METRICS ON TEST SET

| Metric  | Value  |
|---------|--------|
| $R^2$   | 0.9392 |
| RMSE    | 0.2697 |
| MAE (%) | 0.2000 |

### D. Other Tests

Additional tests were performed to understand the capabilities of the model and explore simpler approaches for reliable *AET* prediction.

The primary model employed Principal Component Analysis (PCA) to reduce dimensionality and mitigate multicollinearity, but depends on the availability of the full set of climatological variables. For this reason, a simplified version of the model has been implemented to test if is possible to perform *AET* estimation without all the meteorological variables.

To do so, the correlation matrix (see Figure 12) was used to detect and drop those features having a correlation coefficient that exceeds a defined threshold of  $\pm 0.6$ . In this case mean and maximum air temperature, and maximum relative humidity were excluded, resulting in a final set made of 6 input features. During the testing phase, the simplified model achieved performance comparable to the PCA-based model. The resulting regression plot with 30 days sequences is presented in Figure 15. Additionally, the models have been compared by using shorter input sequences consisting of 14 and 7 consecutive days of meteorological data, and still demonstrate promising results, with a slight drop in the performances.

Moreover, an alternative version of the simplified model was constructed by removing also the precipitation from the input features, since is the most complex and noisy feature to be measured reliably. Also in this case the performances were optimal, demonstrating strong capability to learn local *AET* patterns depending only on temperature, humidity, wind speed and solar radiation.

After having tested the ability of the LSTM of achieving optimal performances even when not all the climatological variables are available, an additional test was performed to evaluate how the model can handle missing values in the feature dataset. In this case, 10% of the values inside the entire features dataset were selected and substituted with the mean value of the feature where their belong, then PCA was applied to remove collinearity. Also this test was conducted with sequences of 30 and 14 days, showing a shrink in the overall precision respect to other models.

Finally, a transferability test was conducted to assess if the model trained over Piedmont region could generalize to a different geographic area. This test was conducted by applying the PCA-based model (30 days) over the Venetian region, located in the north-east of Italy, and results in significantly lower performance, suggesting that the model strongly learns temporal patterns specific to the region.

All the metric values obtained for each test are shown in the table XVI.

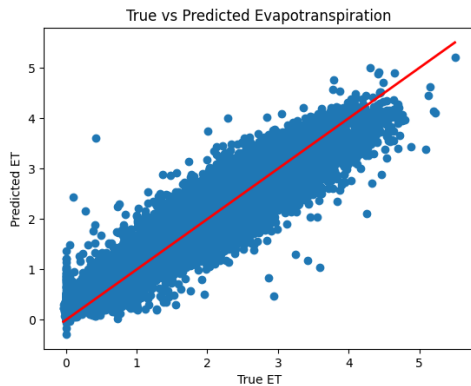


Fig. 15. LSTM Regression Predictions vs. Ground Truth - Reduced Dataset Test

TABLE XVI

EVALUATION METRICS ON TEST SET FOR ALL MODEL CONFIGURATIONS

| Test Configuration                           | $R^2$  | RMSE   | MAE    |
|----------------------------------------------|--------|--------|--------|
| PCA-based model (30 days)                    | 0.9392 | 0.2697 | 0.2000 |
| PCA-based model (14 days)                    | 0.9141 | 0.3212 | 0.2297 |
| PCA-based model (7 days)                     | 0.8556 | 0.4152 | 0.2733 |
| PCA-based, 10% of missing data (30 days)     | 0.8832 | 0.3738 | 0.2578 |
| PCA-based, 10% of missing data (14 days)     | 0.8961 | 0.3532 | 0.2572 |
| Feature-select model (30 days)               | 0.9414 | 0.2648 | 0.1940 |
| Feature-select model (14 days)               | 0.9140 | 0.3213 | 0.2285 |
| Feature-select model, no ppn (30 days)       | 0.9366 | 0.2753 | 0.2023 |
| Feature-select model, no ppn (14 days)       | 0.9115 | 0.3260 | 0.2386 |
| Feature-select model, no ppn (7 days)        | 0.8496 | 0.4238 | 0.2815 |
| Transferability to Venetian region (30 days) | 0.5744 | 0.7125 | 0.4625 |

## VII. CONCLUSIONS

The study presented a comprehensive review that highlighted the most common approaches in the field of evapotranspiration estimation by deep learning. From this review, LSTM appears to be a suitable solution due to its ability to learn seasonal trend and temporal patterns, and remote sensing based and reanalysis datasets emerge as best choices when working with wide study areas.

On these insights, a Long Short-Term Memory (LSTM) model was developed and tested for actual evapotranspiration (AET) prediction in the Piedmont region of Italy, and it demonstrates high accuracy in capturing temporal dependencies, even when input climatological features was missing. The model also showed resilience to missing data, simulating cases where sensors fail to take the measurements.

However, transferability tests indicated that the model's performance significantly deteriorates when applied to a different geographical region, such as Veneto, demonstrating its sensitivity to location-specific climatic patterns.

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